

Output tables for 1xN statistical comparisons.

March 10, 2025

1 Average rankings of Friedman test

Average ranks obtained by each method in the Friedman test.

Algorithm	Ranking
Random	5.6667
NN	4.8333
Sweep	4.8333
SaveSeq	3.1667
SaveParall	1.5
MJ	4.6667
CMT	6.1667
Kilby	5.6667
SaveMatch	9.25
FJ	9.25

Table 1: Average Rankings of the algorithms (Friedman)

Friedman statistic (distributed according to chi-square with 9 degrees of freedom): 33.809091.
P-value computed by Friedman Test: 0.000096.

2 Post hoc comparison (Friedman)

P-values obtained in by applying post hoc methods over the results of Friedman procedure.

i	algorithm	$z = (R_0 - R_i)/SE$	p	Holm	Finner	Li
9	SaveMatch	4.433601	0.000009	0.005556	0.005683	0.034718
8	FJ	4.433601	0.000009	0.00625	0.011334	0.034718
7	CMT	2.669695	0.007592	0.007143	0.016952	0.034718
6	Random	2.383656	0.017142	0.008333	0.022539	0.034718
5	Kilby	2.383656	0.017142	0.01	0.028094	0.034718
4	NN	1.906925	0.05653	0.0125	0.033617	0.034718
3	Sweep	1.906925	0.05653	0.016667	0.039109	0.034718
2	MJ	1.811579	0.070051	0.025	0.04457	0.034718
1	SaveSeq	0.953463	0.340356	0.05	0.05	0.05

Table 2: Post Hoc comparison Table for $\alpha = 0.05$ (FRIEDMAN)

Holm's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.007143 .

Finner's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.033617 .

Li's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.034718 .

3 Adjusted P-Values (Friedman)

Adjusted P-values obtained through the application of the post hoc methods (Friedman).

i	algorithm	unadjusted p	p_{Holm}
1	SaveMatch	0.000009	0.000083
2	FJ	0.000009	0.000083
3	CMT	0.007592	0.053144
4	Random	0.017142	0.10285
5	Kilby	0.017142	0.10285
6	NN	0.05653	0.226121
7	Sweep	0.05653	0.226121
8	MJ	0.070051	0.226121
9	SaveSeq	0.340356	0.340356

Table 3: Adjusted p -values (FRIEDMAN) (I)

i	algorithm	unadjusted p	p_{Finner}	p_{Li}
1	SaveMatch	0.000009	0.000083	0.000014
2	FJ	0.000009	0.000083	0.000014
3	CMT	0.007592	0.022604	0.011378
4	Random	0.017142	0.038156	0.025328
5	Kilby	0.017142	0.038156	0.025328
6	NN	0.05653	0.083586	0.078934
7	Sweep	0.05653	0.083586	0.078934
8	MJ	0.070051	0.083586	0.096001
9	SaveSeq	0.340356	0.340356	0.340356

Table 4: Adjusted p -values (FRIEDMAN) (II)